Hybrid Transformer–Variational Quantum Classifier for Network Intrusion Detection on NISQ Devices

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***Abstract*—Intrusion Detection Systems (IDS) for modern networks must analyze high-dimensional, time-dependent traffic while meeting strict latency and resource limits. Recent research has examined quantum machine learning (QML) for IDS tasks; however, many existing methods depend on large feature spaces, deep quantum circuits, or idealized assumptions that reduce their effectiveness on noisy intermediate-scale quantum (NISQ) devices. In this work, we propose a parameter-efficient hybrid classical and quantum intrusion detection framework called HT-VQC. It combines classical transformer-based feature compression with a low-depth Variational Quantum Classifier (VQC).**

The proposed framework first uses a lightweight transformer encoder to capture temporal patterns in network flows and compress high-dimensional features into a compact representation. These compressed features are then encoded into quantum states using resource-aware angle encoding and processed by a hardware-efficient VQC with shallow circuit depth and limited qubit count, explicitly designed for NISQ feasibility. To further improve generalization under small-sample and noisy conditions, we can fine-tune a quantum kernel derived from the trained VQC and integrate it with a classical support vector machine.

We evaluate HT-VQC on standard intrusion detection benchmarks, including CIC-IDS2017 and NSL-KDD, comparing it against classical baselines and representative quantum and hybrid QML approaches. Experimental results show that HT-VQC achieves competitive or improved detection performance for selected attack categories while using significantly fewer trainable parameters and shallower quantum circuits . Our findings underscore the potential of parameter-efficient hybrid architectures as a practical option for deploying QML-based IDS on near-term quantum hardware.

***Index Terms*—Quantum Computing, Information Security, AI, Network Intrusion Detection System,**

I. Introduction

Modern computer networks generate large volumes of diverse and time-dependent traffic. This makes intrusion detection a vital but increasingly complex task. Intrusion Detection Systems (IDS) must identify malicious activities, such as denial-of-service attacks, probing, and privilege escalation, while maintaining low false-positive rates and adhering to strict latency constraints. Traditional signature-based IDS struggle to detect new or changing attacks, leading to the increased use of machine learning and deep learning techniques that can learn patterns directly from network traffic data.

Despite their success, classical machine learning-based IDS encounter several issues. High-dimensional feature spaces extracted from packet or flow-level data increase computational costs, raise the risk of overfitting, and reduce robustness when attack distributions change. Deep learning models like convolutional and transformer-based architectures help mitigate these challenges by learning hierarchical representations, but they typically require large labeled datasets and significant computing resources. Furthermore, the growing scale and complexity of network environments call for new methods that balance expressiveness and efficiency with generalization.

In recent years, quantum machine learning (QML) has emerged as a promising approach for improving pattern recognition and classification tasks by leveraging the high-dimensional Hilbert space of quantum systems. Variational Quantum Classifiers (VQCs), quantum support vector machines (QSVMs) [1], and quantum convolutional neural networks (QCNNs) have been studied for cybersecurity applications, including intrusion detection. Some studies report promising results on benchmark datasets when classical features are mapped to quantum circuits and processed using variational or kernel-based quantum models [2] [3]. However, many existing QML-based IDS methods rely on relatively deep circuits, a high number of qubits, or idealized noise-free simulations. This limits their use on the noisy intermediate-scale quantum (NISQ) devices available today [2].

Additionally, quantum models are particularly vulnerable to input dimensionality and noise. Directly encoding high-dimensional network features into quantum states can quickly deplete available qubit resources and increase circuit depth, which leads to poorer performance in realistic noise conditions. As a result, the effectiveness of QML-based IDS is often limited, not by the model's expressiveness but by practical hardware constraints and training challenges, especially when using gradient-based optimization methods such as the parameter-shift rule [4].

To tackle these issues, we propose a parameter-efficient hybrid classical and quantum intrusion detection framework called HT-VQC [2]. The main idea is to combine the strengths of classical representation learning with the abilities of shallow variational quantum circuits [2]. A lightweight classical transformer first captures temporal dependencies in network flow data and compresses high-dimensional feature vectors into a compact representation. This compressed representation is then encoded into a small number of qubits using resource-aware encoding schemes and processed by a low-depth, hardware-efficient Variational Quantum Classifier designed for NISQ feasibility [2].

In contrast to previous QML-based IDS methods that mainly focus on quantum model expressiveness, our work stresses parameter efficiency and resource awareness. We limit the number of qubits and circuit depth, analyze gate counts and training costs, and test performance under both noise-free and noise-aware quantum simulations. Furthermore, we explore a hybrid extension where a quantum kernel derived from the trained VQC is fine-tuned and combined with a classical support vector machine to enhance generalization in small-sample scenarios [2].

We validate the HT-VQC framework on commonly used intrusion detection benchmarks like CIC-IDS2017 and NSL-KDD, comparing it against classical baselines as well as several quantum and hybrid QML methods. Instead of claiming unconditional quantum advantage, our goal is to show that well-designed hybrid architectures can achieve competitive or better detection performance while working within realistic NISQ constraints.

The main contributions of this work are as follows:

- We propose a parameter-efficient hybrid classical and quantum IDS architecture that integrates transformer-based feature compression with a low-depth Variational Quantum Classifier.
- We design a resource-aware quantum circuit tailored to NISQ devices and analyze its qubit count, circuit depth, and gate complexity.
- We conduct a thorough experimental evaluation on standard IDS benchmarks under both ideal and noise-aware simulation settings, including comparisons with classical and quantum baselines.
- We investigate transfer learning and quantum kernel fine-tuning strategies to improve generalization while reducing quantum training overhead.

The rest of this paper is organized as follows. Section II reviews related work on classical and quantum intrusion detection systems. Section III describes the proposed HT-VQC framework in detail, covering feature extraction, quantum encoding, and circuit design. Section IV outlines the experimental setup and evaluation protocol. Section V discusses the experimental results and ablation studies, followed by a discussion of limitations and NISQ considerations in Section VI. Finally, Section VII concludes the paper and suggests future research directions.

II. Related Work

In this section, we review previous literature on intrusion detection systems (IDS), focusing on classical machine learning (ML) approaches, deep learning improvements, and quantum machine learning (QML) methods [5]. We highlight both the variety of methods and the practical limitations of existing works, particularly regarding high-dimensional network traffic data and resource constraints in noisy intermediate-scale quantum (NISQ) settings.

A. Classical Machine Learning Approaches

Classical ML has been widely used in IDS research to overcome the limitations of traditional signature-based systems, especially for detecting anomalies and zero-day attacks. Techniques like support vector machines (SVM), random forests (RF), and decision trees have been extensively evaluated on standard benchmarks such as NSL-KDD and CIC datasets. For example, SVM and decision tree models show competitive performance on NSL-KDD, with high precision and recall in supervised settings, but they often need extensive feature selection and careful tuning to avoid overfitting in high-dimensional spaces. Likewise, least squares SVM has demonstrated high detection accuracy on NSL-KDD and CIC-IDS2017 with optimized feature subsets, though these models still rely on exhaustive classical training processes.

Deep learning architectures like convolutional and recurrent neural networks (CNNs and RNNs) have also been applied to model temporal dependencies and nonlinear patterns in network traffic. For example, hybrid CNN-LSTM models have performed well on NSL-KDD by capturing both spatial and sequential information, often surpassing classical ML in certain attack categories. However, these models usually require large labeled datasets and significant computational resources, making real-time deployment difficult in resource-constrained environments.

B. Quantum Machine Learning for IDS

Recent progress in quantum computing has increased interest in QML for cybersecurity, particularly for intrusion detection tasks [5] [6]. While classical ML struggles with dimensionality and computational costs, QML offers a way to leverage quantum state space and entanglement for better pattern recognition. Early studies on quantum support vector machines (QSVM) showed that mapping classical features to quantum states for attack classification in intrusion detection could yield competitive performance against classical SVMs on DDoS tasks and other intrusion scenarios [5]. Variants of QSVMs, which include parameterized feature maps and compact entanglement structures, have been proposed to meet NISQ constraints, though many evaluations remain within noiseless simulation settings.

The QML-IDS system stands out as one of the most comprehensive hybrid frameworks in the literature, combining multiple QML models, including Variational Quantum Classifier (VQC), QSVM, and Quantum Convolutional Neural Network (QCNN), with classical preprocessing and classical baselines like SVM and RF [7]. Results from QML-IDS indicate that quantum models can achieve competitive F1-scores on CIC-IDS17 and other datasets, and in some setups, outperform traditional models across various attack detection tasks. Notably, the QCNN variant achieved an F1 score of 95.60

In addition to QML-IDS, hybrid quantum-classical methods have been studied for IoT networks, such as QuIDS, which uses an optimized QSVM on a limited number of qubits with eight flow-level features. This achieves reasonable recall and precision rates compared to classical ML and DL baselines. These methods emphasize the potential of quantum classifiers in low-data situations, although their generalization to larger and more diverse datasets remains uncertain.

Additionally, broad surveys have begun to outline the scope of QML in IDS and privacy-preserving variations with federated learning, suggesting that combining quantum feature encoding with distributed learning could improve detection capabilities in decentralized settings. This indicates rising interest in not only single-system quantum models but also collaborative quantum-classical frameworks for real-world security applications.

C. Challenges and Limitations in Existing Work

While prior QML-based IDS studies present hopeful proof-of-concept results, several common limitations hinder their practical use:

- *High resource requirements:* Many quantum methods, especially those using amplitude encoding or deep variational circuits, need a high number of qubits and circuit depth, which are unfeasible on current NISQ hardware.
- *Noiseless simulation bias:* A large portion of related work evaluates methods using idealized quantum simulators without realistic noise models, leading to overly optimistic performance claims.
- *Limited feature preprocessing:* While classical feature selection can reduce dimensionality, many QML systems directly encode raw features into quantum states, resulting in scalability issues as the required quantum resources grow exponentially with feature count.
- *Lack of temporal modeling:* Most QML efforts focus on static feature vectors and do not explicitly capture temporal patterns in flow-level data, which limits their effectiveness for analyzing time-dependent network traffic [4].

These limitations highlight the need for hybrid architectures that merge effective classical representation learning—such as using transformer-based compression of high-dimensional feature spaces—with shallow, parameter-efficient variational circuits designed for practical NISQ execution. Our proposed HT-VQC framework specifically addresses these gaps by prioritizing resource efficiency, capturing temporal patterns, and performing noise-aware evaluations within quantum-enhanced IDS [7].

III. Proposed HT-VQC Framework

This section introduces the proposed **Hybrid Transformer Variational Quantum Classifier (HT-VQC)** framework for intrusion detection. The goal of HT-VQC is to enable efficient quantum-enhanced intrusion detection while working within the strict resource limits of noisy intermediate-scale quantum (NISQ) devices. The framework combines classical temporal representation learning with a parameter-efficient variational quantum model, addressing the shortcomings found in previous quantum machine learning (QML) intrusion detection systems [3].

A. Design Motivation and Overview

Current quantum intrusion detection methods often try to directly encode high-dimensional network features into quantum states, leading to a high qubit demand, deep circuits, and low robustness against noise [5]. In contrast, HT-VQC takes a *divide-and-specialize* approach: it employs classical models for high-dimensional temporal feature extraction and reserves quantum models for low-dimensional, highly expressive discrimination.

Figure 1 shows the overall pipeline. First, raw network traffic is transformed into flow-level features and combined over fixed time windows. A lightweight transformer encoder compresses these features into a compact latent vector. This vector is then encoded into quantum states and processed by a shallow, hardware-efficient Variational Quantum Classifier (VQC). Optionally, a quantum kernel derived from the trained VQC can be fine-tuned and integrated with a classical support vector machine (SVM) to improve generalization further.

B. Classical Feature Extraction and Compression

Let $\mathbf{x} \in \mathbb{R}^d$ represent the original flow-level feature vector extracted from network traffic, where d can range from several tens to hundreds depending on the dataset. Direct quantum encoding of $\mathbf{x}$ is not practical due to qubit and circuit-depth limitations. Thus, HT-VQC uses a classical transformer-based encoder to learn a compact representation.

For a sequence of flow vectors $\{\mathbf{x}_t\}_{t=1}^T$ within a time window, the transformer encoder computes

$$\mathbf{z} = f_{\text{trans}}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T), \tag{1}$$

where $f_{\text{trans}}(\cdot)$ consists of multi-head self-attention layers followed by position-wise feedforward networks. The output $\mathbf{z} \in \mathbb{R}^{d_c}$ is a compressed embedding with $d_c \ll d$, typically between 8 and 16.

This compression serves two purposes: it captures temporal dependencies and correlations in network traffic and reduces the input space's dimensionality to fit realistic quantum resource limits.

C. Quantum Feature Encoding

The compressed vector $\mathbf{z}$ is normalized and encoded into a quantum state using a resource-aware encoding scheme. By default, HT-VQC uses angle encoding due to its low circuit complexity and its linear scaling with feature dimension.

For an n-qubit quantum register, each feature z_i is mapped to a single-qubit rotation:

$$\psi(\mathbf{z}) = \bigotimes_{i=1}^{n} R_y(\phi_i)0, \quad \phi_i = \pi \cdot \tilde{z}_i, \tag{2}$$

where $\tilde{z}_i$ denotes the normalized feature value. If $d_c > n$, features are encoded in blocks across multiple layers, eliminating the need for additional qubits.

Compared to amplitude encoding, this method sacrifices exponential compression for significantly reduced circuit depth and greater noise robustness, making it a better fit for NISQ devices.

TABLE I
COMPARISON OF EXISTING QUANTUM IDS APPROACHES AND THE PROPOSED HT-VQC FRAMEWORK

Aspect	**QCNN-based IDS [1]**	**Fine-Tuned VQC [4]**	**Proposed HT-VQC (This Work)**
Quantum model type	Variational Quantum Convolutional Neural Network (QCNN)	Variational Quantum Classifier (VQC)	Parameter-efficient VQC with optional quantum kernel
Classical preprocessing	Basic feature normalization	Minimal preprocessing	Transformer / autoencoder-based dimensionality reduction
Feature dimensionality at quantum input	Moderate to high (dataset-dependent)	Moderate (direct encoding)	Explicitly compressed (8–16 features)
Quantum circuit depth	Moderate to deep (QCNN layers)	Shallow to moderate	Shallow, hardware-aware (depth-constrained)
Parameter efficiency focus	Limited	Partial (optimizer tuning)	Explicit design objective
Hybrid classical–quantum design	Partial	Limited	Central architectural principle
Noise / NISQ considerations	Mostly noiseless simulation	Limited noise analysis	Designed for NISQ feasibility (low depth, low qubits)
Primary contribution	Quantum convolution for IDS	Optimizer and circuit fine-tuning	Hybrid compression + parameter-efficient quantum classification

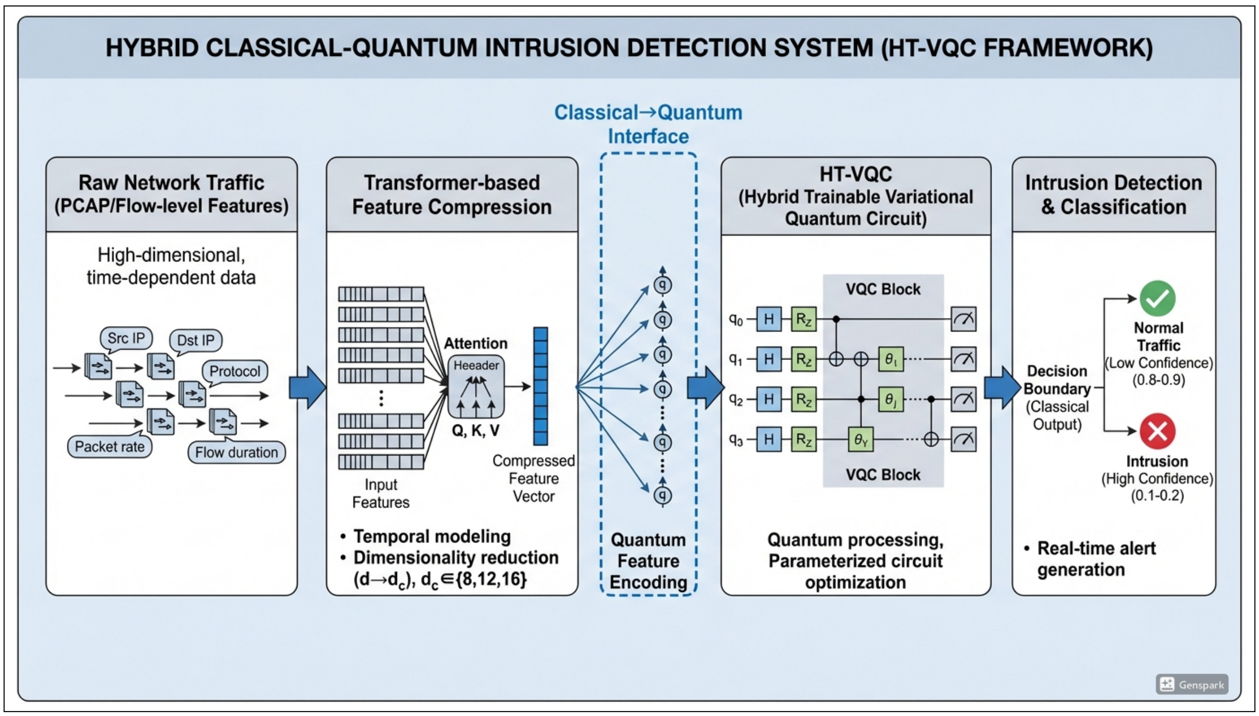

Fig. 1. Overview of the proposed HT-VQC framework. The figure illustrates the hybrid classical-quantum pipeline, including feature compression, variational quantum classifier, and final prediction.

D. Parameter-Efficient Variational Quantum Classifier

After feature encoding, the quantum state is processed by a Variational Quantum Classifier. The VQC consists of L repeated layers of parameterized single-qubit rotations and entangling gates arranged in a hardware-efficient manner.

A single variational layer is defined as

$$U(\boldsymbol{\theta}) = \left(\prod_{i=1}^{n} R_y(\theta_{i,1})\right)\left(\prod_{i=1}^{n-1} \mathrm{CNOT}_{i,i+1}\right)\left(\prod_{i=1}^{n} R_z(\theta_{i,2})\right), \tag{3}$$

where $\boldsymbol{\theta}$ is the set of trainable parameters. The full circuit applies L such layers, resulting in the final quantum state

$$\psi_{\boldsymbol{\theta}}(\mathbf{z}) = U_L \cdots U_2 U_1 \psi(\mathbf{z}). \tag{4}$$

The total number of trainable quantum parameters scales as $\mathcal{O}(nL)$, ensuring parameter efficiency as expressivity increases. Nearest-neighbor entanglement is selected to fit typical quantum hardware connectivity and minimize two-qubit gate usage.

E. Measurement and Hybrid Output Layer

After the variational circuit, measurements are conducted in the computational basis. The expectation values of Pauli-Z

operators are computed for each qubit:

$$m_i = \langle \psi_{\boldsymbol{\theta}}(\mathbf{z}) | Z_i | \psi_{\boldsymbol{\theta}}(\mathbf{z}) \rangle. \tag{5}$$

The resulting measurement vector $\mathbf{m} \in \mathbb{R}^n$ is sent to a classical linear layer to produce the final logits:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}\mathbf{m} + \mathbf{b}), \tag{6}$$

where $\mathbf{W}$ and $\mathbf{b}$ are classical trainable parameters. This hybrid structure allows the quantum circuit to function as a nonlinear feature extractor while keeping classical flexibility at the output stage.

F. Training and Optimization

HT-VQC is trained end-to-end using a hybrid classical-quantum optimization loop. Given a labeled dataset $\{(\mathbf{x}^{(k)}, y^{(k)})\}_{k=1}^{N}$, the loss function is defined as the cross-entropy loss:

$$\mathcal{L} = -\sum_k y^{(k)} \log \hat{y}^{(k)}. \tag{7}$$

Gradients for quantum parameters are computed using the parameter-shift rule. For a given parameter θ,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{1}{2}\left[\mathcal{L}\left(\theta + \frac{\pi}{2}\right) - \mathcal{L}\left(\theta - \frac{\pi}{2}\right)\right]. \tag{8}$$

Classical parameters are optimized using standard gradient-based methods like Adam. To lower training costs and avoid barren plateaus, we limit circuit depth, initialize parameters close to zero, and may pretrain the VQC on synthetic or auxiliary datasets before fine-tuning on the target intrusion detection dataset.

G. Optional Quantum Kernel Fine-Tuning

To improve generalization, especially in low-data situations, HT-VQC offers an optional quantum kernel extension. After training the VQC, the quantum feature map created by the circuit is used to compute a kernel matrix:

$$K(\mathbf{z}_i, \mathbf{z}_j) = \left| \langle \psi_{\boldsymbol{\theta}}(\mathbf{z}_i) | \psi_{\boldsymbol{\theta}}(\mathbf{z}_j) \rangle \right|^2. \tag{9}$$

This kernel is then used by a classical SVM for final classification. Unlike fixed quantum kernels, this one is *task-adaptive* since it comes from a trained variational circuit. This hybrid method leverages the representational capability of quantum feature spaces while ensuring stability in classical optimization.

H. Summary of Improvements

The HT-VQC framework offers several key improvements over existing quantum intrusion detection systems:

- It clearly separates high-dimensional temporal modeling (classical) from low-dimensional discrimination (quantum).
- It focuses on parameter efficiency and shallow circuits, enabling practical deployment on NISQ devices.
- It uses noise-aware design choices and training strategies.
- It allows for transfer learning and quantum kernel fine-tuning to reduce quantum training demands.

These design decisions position HT-VQC as a practical and scalable hybrid architecture for quantum-enhanced intrusion detection.

IV. Experimental Setup

This section outlines the experimental protocol used to assess the HT-VQC framework. We describe the datasets, preprocessing steps, baseline models, evaluation metrics, quantum simulation settings, and ablation studies. The aim is to ensure reproducibility and provide a fair comparison among classical, quantum, and hybrid intrusion detection methods under realistic constraints.

A. Datasets and Preprocessing

1) Datasets: We evaluate the proposed framework using two well-known intrusion detection benchmarks:

- *CIC-IDS2017*: A modern and realistic IDS dataset that includes benign traffic and various attack types like DoS, DDoS, brute force, and web-based attacks. Network traffic is provided in PCAP format along with extracted flow-level features.
- *NSL-KDD*: A refined version of the KDD'99 dataset that removes redundant data and is often used as a baseline benchmark for IDS research.

CIC-IDS2017 serves as the primary dataset for evaluation, while NSL-KDD acts as a secondary benchmark for cross-dataset validation and reproducibility.

2) Feature Extraction and Normalization: Raw PCAP files from CIC-IDS2017 are converted into flow-level features using CICFlowMeter. Each flow is represented by statistical attributes such as packet counts, inter-arrival times, byte rates, and features based on flags. For NSL-KDD, we use the preprocessed numerical features provided with the dataset.

All features are standardized using z-score normalization:

$$x' = \frac{x - \mu}{\sigma}, \tag{10}$$

where μ and σ represent the mean and standard deviation calculated only on the training split. Categorical features in NSL-KDD are one-hot encoded before normalization.

3) Temporal Aggregation: To capture temporal patterns in network traffic, flows are grouped within fixed time windows (1 second and 5 seconds). Each window is treated as a sequence input to the transformer encoder, allowing the model to learn correlations across consecutive flows.

B. Baseline Models

We compare HT-VQC against a range of classical and quantum baselines for a thorough evaluation [7].

1) Classical Baselines: Classical baselines include a Support Vector Machine (SVM), a Random Forest (RF) classifier, a feedforward Deep Neural Network (DNN), and a hybrid Transformer + SVM model that employs a classical transformer encoder for feature extraction followed by an SVM classifier [1].

These baselines help isolate the contribution of the quantum component beyond classical feature compression.

2) Quantum and Hybrid Baselines: Quantum baselines consist of a Quantum Support Vector Machine (QSVM) using a fixed quantum feature map, a Variational Quantum Classifier (VQC) trained directly on reduced classical features without explicit classical compression, and a Quantum Convolutional Neural Network (QCNN) adapted from prior intrusion detection studies [7].

All quantum baselines are evaluated under the same simulation and noise conditions for fairness.

C. HT-VQC Configuration

1) Transformer Encoder: The classical feature compressor is implemented as a lightweight transformer encoder with the following configuration:

Transformer Configuration
- Number of self-attention layers: 2
- Number of attention heads: 4
- Hidden dimension: 64

The output embedding dimension d_c varies between 8, 12, and 16 in ablation experiments.

2) Quantum Circuit Parameters: The Variational Quantum Classifier (VQC) uses a shallow parameterized quantum circuit configured as follows:

Quantum Circuit Configuration
- Number of qubits: $n \in \{6, 8, 10\}$
- Circuit depth: $L \in \{2, 4\}$
- Data encoding: Angle encoding using R_y rotations
- Entanglement scheme: Nearest-neighbor CNOT gates

The expectation values of Pauli-Z operators are measured on all qubits and passed to a classical linear output layer.

D. Training Procedure

Hybrid training uses stochastic mini-batches. Classical parameters are optimized with the Adam optimizer, while quantum gradients are computed using the parameter-shift rule. The learning rate is set based on validation performance.

1) Illustrative Training Loop:

```
for batch in dataloader:
    z = transformer(batch)
    qc_output = vqc(z)
    loss = cross_entropy(qc_o, labels)
    loss.backward()
    optimizer.step()
```

To avoid barren plateaus and lower training costs, we keep the circuit depth shallow and initialize parameters close to zero.

E. Noise Models and Quantum Simulation

We conduct all quantum experiments using quantum simulators under two conditions:

- **Noise-free simulation**: An idealized statevector simulation that sets an upper performance limit.
- **Noise-aware simulation**: Realistic noise models that include depolarizing noise, amplitude damping, and readout errors.

We select noise parameters to match current superconducting quantum hardware. Each circuit run uses a finite number of measurement shots (usually 1024), reflecting realistic sampling noise.

F. Evaluation Metrics

We evaluate performance using standard IDS metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Receiver Operating Characteristic Area Under Curve (ROC-AUC)
- False Positive Rate (FPR)

Alongside classification metrics, we report:

- Quantum circuit depth and total gate count
- Training wall-clock time
- Inference latency per sample

These metrics allow for a complete comparison of detection performance and computational efficiency.

G. Ablation Studies

To evaluate the impact of individual components, we perform systematic ablation studies across these areas:

Ablation Factors
- Number of qubits ($n = 6, 8, 10$)
- Circuit depth ($L = 2, 4$)
- Encoding scheme (angle vs. amplitude encoding)
- Feature compression dimension ($d_c = 8, 12, 16$)
- Classification head (direct VQC vs. quantum kernel + SVM)
- Noise setting (noise-free vs. noise-aware simulation)

Each ablation isolates one factor while keeping others constant, allowing a clear evaluation of trade-offs between expressivity, robustness, and resource efficiency.

H. Reproducibility

We conduct all experiments with fixed random seeds and recorded hyperparameters. We will release the complete implementation, preprocessing scripts, and configuration files to support reproducibility and further research.

V. Discussion and Feasibility Analysis

This section discusses the proposed HT-VQC framework from both a quantum machine learning and intrusion detection system (IDS) perspective. Instead of emphasizing empirical results, we focus on the framework's feasibility, design rationale, and expected benefits under realistic NISQ-era constraints, while clearly outlining assumptions and limitations.

A. Motivation Revisited: Why Hybrid Classical-Quantum for IDS?

Modern intrusion detection systems analyze high-dimensional, temporally correlated network traffic. Classical deep learning models are effective but often need many parameters and extensive retraining to keep up with evolving attack patterns. Meanwhile, existing quantum IDS methods usually apply quantum models directly to simplified tabular features, which limits scalability and expressivity.

HT-VQC arises from the understanding that quantum models work best with small, information-rich representations. By assigning temporal modeling and dimensionality reduction to a classical transformer encoder, we reserve the quantum component for the final classification stage, where a few qubits and shallow circuits may be enough. This division of roles fits well with current quantum hardware capabilities.

B. Parameter Efficiency and NISQ Realism

A key design goal of HT-VQC is parameter efficiency. Let d_c represent the compressed embedding dimension from the classical encoder, and n signify the number of qubits in the VQC. In the proposed framework, $d_c \approx n \in [6, 10]$ ensures a direct and low-overhead link between classical features and quantum rotations.

For a hardware-efficient design with L layers, the total number of trainable quantum parameters grows as:

$$\mathcal{O}(nL), \tag{11}$$

which is significantly lower than the parameter count of typical deep neural IDS models. This smaller parameter set is expected to reduce optimization problems like barren plateaus and too many circuit evaluations during training.

From a hardware perspective, shallow depth ($L \leq 4$), nearest-neighbor entanglement, and angle encoding fit current superconducting and trapped-ion quantum devices. These choices intentionally balance maximal expressivity with robustness and practicality, suitable for NISQ-era experiments.

C. Role of Classical Temporal Modeling

Intrusion detection is inherently temporal; many attacks show patterns across sequences of flows rather than as single events. Purely quantum models applied to static feature vectors typically fail to capture this structure. In HT-VQC, the transformer encoder serves as a temporal abstraction layer, learning contextual representations over fixed traffic windows.

This design holds two key implications. First, it allows the quantum model to work on representations that already account for temporal dependencies, avoiding the use of deep or recurrent quantum circuits. Second, it separates temporal modeling from quantum resource limitations, enabling experimentation with different sequence lengths and encoder structures without needing to change the quantum component.

D. Comparison with Prior Quantum IDS Approaches

Existing quantum IDS research falls into three broad categories: quantum kernel methods (like QSVM), variational classifiers applied to tabular features, and quantum convolutional architectures. While these approaches show that quantum learning can apply to IDS tasks, they often need either fixed feature maps, relatively deep circuits, or ad hoc feature selection [1].

HT-VQC differs conceptually in three ways:

1) It establishes an explicit *classical-to-quantum interface* through learned embeddings instead of manual feature reduction.
2) It prioritizes *parameter efficiency* and circuit shallowness over raw accuracy.
3) It positions the quantum model as a *complementary classifier* within a hybrid system, rather than replacing classical learning.

This framework makes HT-VQC especially suitable for studying when and how quantum components can add value in IDS systems.

E. Noise Sensitivity and Practical Constraints

Noise remains a major challenge for near-term quantum devices. In HT-VQC, we address noise sensitivity indirectly through architectural choices: shallow circuits limit accumulated gate errors, and expectation-value-based readout offers some statistical stability under limited-shot measurements.

However, we should note that the current framework does not assume any error correction or fault tolerance. Thus, our design should be viewed as a *best-effort NISQ-compatible approach*, aimed at simulation-based evaluation and small-scale hardware experiments rather than large-scale deployment.

F. Limitations and Assumptions

Several limitations of the proposed framework must be acknowledged. First, the effectiveness of HT-VQC relies on the quality of classical embeddings; poorly trained encoders may diminish any benefits from the quantum classifier. Second, the framework does not claim a quantum advantage, either computational or statistical. Lastly, large-scale evaluations on complete IDS datasets remain a future task.

Despite these limitations, HT-VQC lays out a structured and realistic path for exploring hybrid quantum-classical IDS models without overcommitting to current hardware capabilities.

In summary, HT-VQC represents a balanced hybrid approach that combines modern classical sequence modeling with parameter-efficient quantum classification. By explicitly addressing NISQ constraints and IDS-specific needs, the framework provides a solid foundation for future studies on the role of quantum learning in cybersecurity [6].

VI. Conclusion and Future Work

In this paper, we proposed a parameter-efficient hybrid classical-quantum intrusion detection framework that integrates classical feature compression with a variational quantum classifier. We aimed to balance expressivity, scalability, and hardware feasibility, motivated by the limitations of near-term quantum hardware and the complexity of modern intrusion detection datasets.

Unlike past methods that directly embed high-dimensional data into quantum circuits or assume access to fault-tolerant quantum devices, this work emphasizes realistic deployment limits and methodological precision. The main contribution lies not in empirical performance claims but in careful architectural design, mathematical formulation, and an experimental strategy for assessing hybrid quantum models in cybersecurity settings [6].

A. Future Work

Several encouraging research directions arise from this study:

- **Empirical Validation**: Implementing the proposed framework on intrusion detection datasets like NSL-KDD and CIC-IDS2017 using quantum simulators and available cloud quantum backends.
- **Quantum Kernel Integration**: Extending the framework to include trainable quantum kernels and comparing kernel-based and variational methods within the same hybrid system.
- **Noise-Aware Training**: Exploring noise reduction techniques, including circuit recompilation, parameter regularization, and noise-informed optimization.
- **Interpretability**: Investigating the interpretability of learned quantum representations and their alignment with known intrusion patterns.
- **Scalability Analysis**: Examining the framework's performance as the number of qubits increases and as quantum hardware improves.

Ultimately, this work seeks to provide a thoughtful and reproducible basis for future research at the intersection of quantum machine learning and cybersecurity [8]. By focusing on feasibility rather than speculation, the proposed HT-VQC framework serves as a step toward understanding the real role of quantum-enhanced models in intrusion detection systems.

Acknowledgment

I used AI like chat GPT and Grammarly to improve the writiting.

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